Udit Ramawat

Technical Program Leader | AI Products, Aviation Software, Frontier Hardware
ramawat.fyi

Summary

Proven technical program leader specializing in AI-native products, aviation software, spatial computing, and frontier hardware/software systems. Leads cross-functional execution from ambiguous research, architecture, and operational needs through integrated plans, requirements, release readiness, and delivery. Proven record aligning 60+ engineering teams and 120+ stakeholders across AI/ML, application software, and hardware lifecycles. Hands-on builder who uses AI coding agents to prototype internal tools and workflow automations that accelerate engineering execution.

Experience

Archer Aviation San Jose, CA

Sr. Staff Technical Program Manager *July 2025 - Present*

- Standardized ASR evaluation across Data Engineering and ML Research, defining the eval contract and moving iOS SDK validation from local runs to concurrent device-fleet CI.
- Scaled CI/CD reliability across iOS and backend by expanding test coverage and automating vulnerability detection, dependency triage, and remediation.
- Built a full-stack product insights platform that correlates releases with customer feedback across Slack, Google Forms, and Linear to surface sentiment, regressions, and emerging themes.
- Built and shipped an AI-assisted flight-test application for dynamic signal thresholds and automated alerts, enabling a previously unavailable engineering workflow.
- Lead roadmap and execution for Archer Intelligence, translating AI/ML research, engineering, and operational needs into prioritized product investments.

Apple Cupertino, CA

Staff Engineering Program Manager, Vision Products Group *Oct 2024 - July 2025*

- Led product definition for real-time platform software architecture on next-generation spatial computing systems, spanning machine perception, sensor-driven experiences, and on-device intelligence.
- Aligned roadmap dependencies across 60+ engineering teams to de-risk delivery of launch-critical platform capabilities for future hardware.
- Defined product direction for on-device ML and LLM optimizations within strict thermal, latency, and power constraints.
- Drove product planning for cross-platform machine-perception capabilities, integrating novel sensor inputs into spatial computing experiences.

Senior Engineering Program Manager, AI/ML *Oct 2023 - Oct 2024*

- Led product strategy for device-integrated Siri and AI/ML experiences across Home Devices and unreleased hardware platforms, focused on latency, reliability, and architectural readiness.
- Defined product and technical direction for Siri on unreleased hardware, unifying requirements across 30+ engineering teams for Day-1 readiness.
- Established a cross-functional operating model for 120+ stakeholders across Engineering, Design, Marketing, and leadership, improving dependency visibility and feature execution.
- Translated AI/ML, latency, privacy, and platform constraints into product requirements, roadmap tradeoffs, and launch decisions.

Senior Engineering Program Manager, Vision Products Group *Oct 2021 - Oct 2023*

- Led product strategy and execution for pre-launch Apple Vision Pro platform software capabilities, framework migration, software update/recovery flows, and service readiness.
- Defined roadmap, success metrics, and readiness mechanisms for low-level platform capabilities, improving visibility into adoption, activation, quality, and launch risk.
- Drove migration planning for key visionOS frameworks and APIs ahead of launch, aligning engineering owners, readiness criteria, and adoption milestones.
- Led product definition for visionOS software update and recovery mechanisms, partnering with Design and Engineering to reduce user friction and improve serviceability.
- Established operational processes for AppleCare and factory repair workflows, enabling service readiness for a brand-new product category.

Engineering Program Manager, Software Engineering *Nov 2019 - Oct 2021*

- Led cross-functional prioritization for factory software supporting Apple Silicon Macs, Magic Keyboard, and AirPods from concept through launch.
- Drove readiness for core security infrastructure across iOS, macOS, watchOS, and visionOS manufacturing lines, helping scale reliable production workflows.
- Built automated reporting pipelines to identify, triage, and escalate software blockers across factory lines, reducing time to resolution for critical manufacturing issues.
- Partnered with software engineering, operations, security, and manufacturing teams to translate production constraints into roadmap priorities and launch readiness plans.

Intel Chandler, AZ

Product Manager, Data Protection *Mar 2016 - Nov 2019*

- Led product delivery for enterprise data protection solutions spanning data classification, DLP, and encryption workflows.
- Led a cross-functional team of 10+ developers and system engineers, driving 54% year-over-year growth in requests serviced by the encryption infrastructure.
- Released multiple versions of proprietary SDKs that expanded automation adoption across enterprise environments.

Apttus San Mateo, CA

Salesforce Analyst *Nov 2015 - Mar 2016*

- Provided premium support to high-value customers by debugging complex technical issues related to CPQ and CLM offerings.
- Developed troubleshooting guides and process articles to accelerate customer implementations.

Education

Santa Clara University Santa Clara, CA

MS, Computer Science & Engineering

Illinois Institute of Technology Chicago, IL

BS, Computer Science